SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CAPSTONE PRESENTATION 1

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 3 – CAPSTONE 1
Weightage:	30 % of CIA	**Total Marks:**	100 MARKS
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	C Sangita	**Reg. No.:**	192373059
Section / Batch:	Slot-A/2023	**Date:**	16-10-2026

Code of Conduct

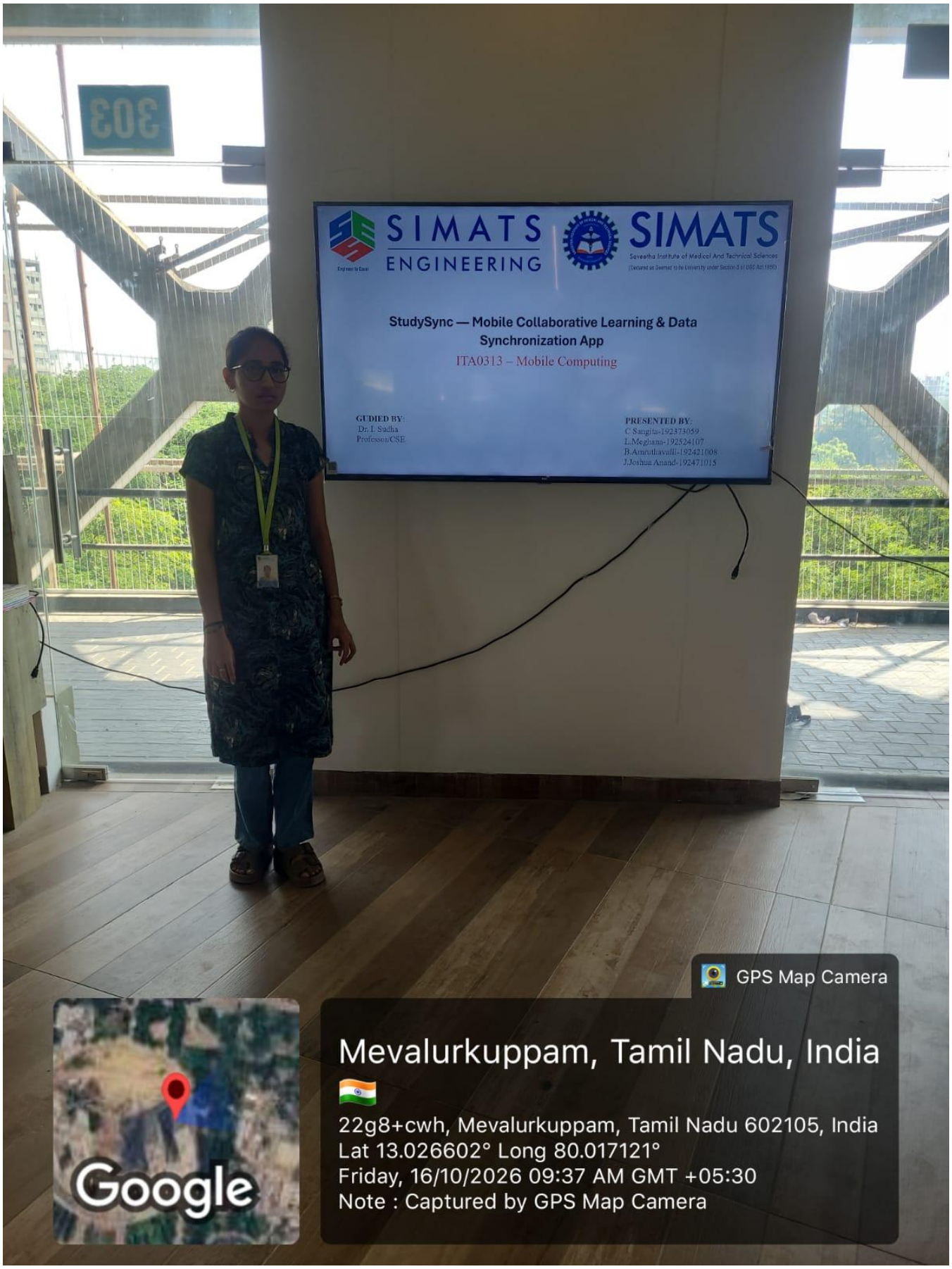
I C Sangita(192373059) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: sangita.c

Instructions

- Duration: 10 minutes.
- Attach ppt in handout manner
- Attach photo of your presentation

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____


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StudySync — Mobile Collaborative Learning & Data Synchronization App
ITA0313 – Mobile Computing
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GPS Map Camera
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22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026602° Long 80.017121°
Friday, 16/10/2026 09:37 AM GMT +05:30
Note : Captured by GPS Map Camera

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StudySync — Mobile Collaborative Learning & Data Synchronization App

ITA0313 – Mobile Computing

GUDIED BY:
Dr. I. Sudha
Profess/CSE

PRESENTED BY:
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Introduction

- StudySync is a mobile-based collaborative learning application designed to make learning resources accessible and organized.
- It allows students to create profiles, access study materials, share notes, and exchange knowledge with peers.
- The application supports collaborative activities such as group discussions, shared notes, and learning resource sharing.
- A data synchronization mechanism helps maintain consistent learning information across multiple devices.
- Learning analytics help students track their learning activities and progress.
- Notifications provide updates about shared resources, group activities, and synchronization status.

Problem Statement

- Students often use multiple applications and devices to access and manage their study materials.
- Sharing notes and learning resources between students can be unorganized and time-consuming.
- Changes made to shared content may not appear immediately on other devices.
- Poor network connectivity can interrupt access to recently updated learning materials.
- Conflicting edits and duplicate content can create inconsistencies in collaborative learning.
- Students may lack a centralized system to monitor their learning activities and receive relevant updates.
- To develop a mobile collaborative learning application that simplifies resource management, supports knowledge exchange, maintains data consistency across devices, and provides learning analytics and timely notifications.

Objectives

- Identity and Resource Management: Provide secure user authentication, profiles, and organized learning resources.
- Collaborative Learning: Enable students to share notes, exchange knowledge, and collaborate with peers.
- Data Synchronization: Keep learning data consistent across devices and handle offline changes.
- Conflict Management: Identify conflicting updates and apply appropriate consistency rules.
- Learning Analytics: Track learning activities and present useful progress information.
- Notification Management: Notify users about shared content, collaboration updates, and synchronization events.

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Literature Review

S.No	Area	Existing Method	Limitation	Our Project
Mobile Learning	Mobile learning applications	Limited integration of collaboration and synchronization	Provides a unified mobile learning platform	Mobile Learning
Collaborative Learning	Group discussions and shared documents	Difficulty managing shared resources and updates	Enables organized knowledge exchange	Collaborative Learning
Data Synchronization	Cloud-based synchronization	Conflicts may occur during offline edits	Supports conflict detection and consistency management	Data Synchronization
Learning Analytics	Activity tracking and progress reports	Limited integration with collaborative activities	Combines learning analytics with notifications	Learning Analytics

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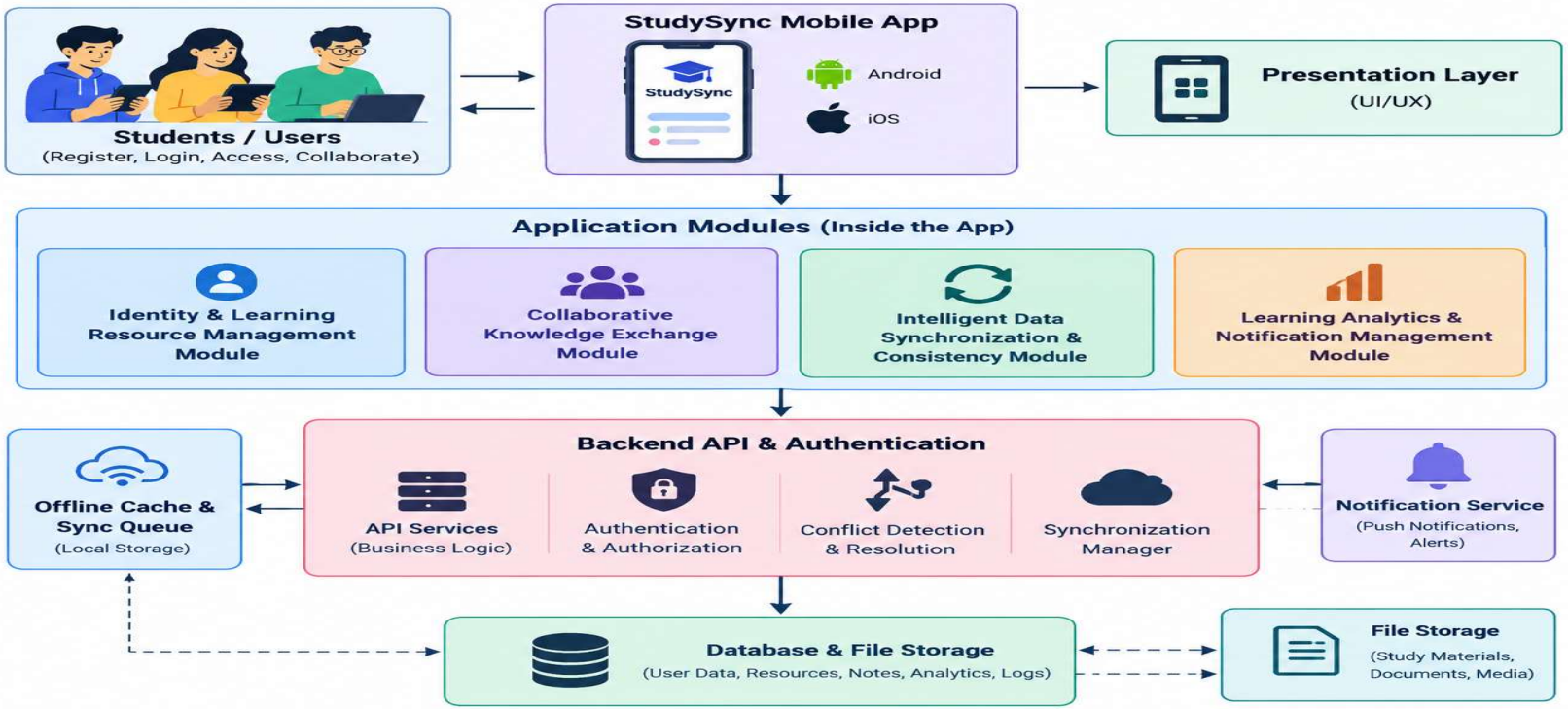
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System Architecture

StudySync — Mobile Collaborative Learning & Data Synchronization App



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Module 1: Identity & Learning Resource

This module provides a centralized platform for managing student identities, controlling access, and organizing educational resources for easy retrieval and secure sharing.

- User registration and login with secure authentication.
- Student profile creation and profile information management.
- Role-based access for students and administrators.
- Uploading, downloading, and managing study materials.
- Categorizing learning resources based on subjects and topics.
- Searching and retrieving notes, documents, and learning links.
- Managing resource permissions and authorized access.
- Maintaining resource details and availability for collaborative learning.

Module 2: Collaborative Knowledge Exchange

This module creates a collaborative learning environment where students can exchange ideas, share educational materials, and participate in group-based learning activities.

- Creating and joining subject-based study groups.
- Sharing notes, documents, presentations, and learning links.
- Discussing academic topics and exchanging ideas.
- Posting questions and sharing answers with group members.
- Adding comments and feedback to shared learning resources.
- Viewing updates and contributions from other students.
- Supporting peer-to-peer knowledge sharing and group learning.
- Maintaining organized shared resources for easy access by group members.

Module 3: Intelligent Data Synchronization & Consistency

This module maintains consistent learning information across multiple devices by synchronizing updates, supporting offline access, and managing conflicting changes.

- Automatically synchronizing learning resources across connected devices.
- Storing selected learning materials locally for offline access.
- Maintaining a queue of changes made during network disconnection.
- Detecting conflicting updates when multiple users modify shared content.
- Using timestamps and version numbers to identify changes.
- Applying conflict-resolution rules to maintain data consistency.
- Retrying failed synchronization operations when connectivity returns.
- Displaying synchronization status and ensuring pending updates are processed.

Module 4: Learning Analytics & Notification Management

This module monitors learning activities, presents student progress information, and delivers timely notifications to improve engagement and collaboration.

- Tracking access to study materials and shared resources.
- Monitoring participation in collaborative learning groups.
- Recording relevant learning activities and resource interactions.
- Displaying learning activity summaries through dashboards.
- Generating progress reports based on recorded learning activities.
- Sending notifications about newly shared materials and group updates.
- Providing alerts for synchronization success, failure, and pending updates.
- Supporting activity reminders and personalized notification preferences.

Expected Outcomes

- StudySync is expected to provide an integrated mobile learning platform that improves collaboration, simplifies learning resource management, and maintains consistent data across multiple devices.
- Centralized management of student profiles and learning resources.
- Improved knowledge sharing and peer-to-peer collaboration.
- Reliable data synchronization across devices with conflict resolution.
- Offline access to learning materials and automatic updates when connectivity is restored.
- Better understanding of student learning activities through analytics dashboards.
- Timely notifications about shared resources, group discussions, and synchronization updates.
- Enhanced accessibility, organization, and overall learning experience.

Sustainable Development Goals (SDGs)

- **SDG 4 – Quality Education**

Provides easy access to digital learning resources.
Promotes collaborative learning and knowledge sharing.
Supports inclusive and accessible learning opportunities.

- **SDG 9 – Industry, Innovation and Infrastructure**

Uses mobile technology to support digital education.
Improves learning resource availability through data synchronization.
Encourages innovative technology-based learning solutions.

- **SDG 10 – Reduced Inequalities**

Helps students access shared educational resources.
Supports learning across different locations.
Encourages equal opportunities for knowledge sharing.



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Thank You